

1. INTRODUCTION

This report extends the multimodal emotion recognition and sentiment analysis pipeline built and reproduced in Assignments 1 and 2, respectively. The baseline system in the original paper by Farhadipour et al. (2025) uses text (RoBERTa), audio (Wav2Vec2), facial expressions (FacialNet) and video (CNN+Transformer) to predict emotion and sentiment on the MELD dataset. Assignment 2 reproduced that system on a class-balanced 15% subset of MELD, and replicated the finding that text is the primary contribution to multimodal emotion recognition, while the visual modalities are less consistent and reliable.

Assignment 3 extends those baselines in two ways. First, we replace the feature concatenation in the multimodal fusion stage with a gated attention fusion model that learns sample-wise weights for each modality to allow the classifier to rely on more confident modalities at test time. Second, we present additional results on IEMOCAP and perform domain transfer experiments, we pretrain a RoBERTa text model on IEMOCAP and test it on MELD zero-shot and fine-tuned for both the emotion and sentiment tasks.

2. PAPER SUMMARY

Farhadipour et al. (2025) explore emotion recognition and sentiment analysis in multi-party dialogues, using the MELD dataset from the Friends TV show. It comprises over 13,000 utterances labeled with seven emotions (anger, disgust, fear, joy, neutral, sadness, surprise) and three sentiments (positive, negative, neutral). Unlike most existing studies using the clean IEMOCAP data, MELD is a more practical dataset with background noise, multi-speaker scenarios and changing camera perspectives, making it a more difficult task.

The proposed system has four parallel streams for different modalities. For text, a RoBERTa model is first pretrained on 58 million tweets through the TweetEval benchmark, and then fine-tuned on MELD. For audio, Wav2Vec2 is pretrained on several speech databases containing emotional expressions and then fine-tuned on MELD. For faces, a pipeline called FacialNet uses TalkNet to identify the active speaker, InfoMap clustering to extract the target speaker's face, InceptionResNetV1 for feature extraction and a Bidirectional LSTM with attention for temporal classification. For video, a CNN+Transformer model has been designed with MobileNetV2 for spatial feature extraction and a local self-attention transformer for temporal feature extraction.

The feature embeddings from the four pipelines are combined into a multimodal vector and passed to a Multi-Layer Perceptron. The paper claims that the use of all four modalities achieves 66.36% for emotion recognition and 72.15% for sentiment analysis on the test set, with improvements of 2.02% and 2.94% respectively over the best unimodal baseline.

3. REPRODUCTION SUMMARY

In Assignment 2, we re-implemented the four unimodal pipelines and experimented with multimodal fusion on a 15% balanced sub-sample of MELD. The computing platform was Kaggle with 2 NVIDIA Tesla T4 GPUs (15 GB VRAM each), 30 GB RAM, and a time limit of 12 hours. The main changes from the original paper were: use of roberta-base instead of the TweetEval-pretrained model, use of MTCNN for face detection instead of TalkNet+InfoMap, and use of global self-attention instead of local attention in the video transformer.

3.1. Unimodal Results from Assignment 2

Modality	Train Acc (%)	Dev Acc (%)	Test Acc (%)	Paper Acc (%)
Text – RoBERTa-base	81.87	60.12	60.93	64.34
Audio – Wav2Vec2-base	47.19	42.94	48.33	51.49

MULTIMODAL EMOTION RECOGNITION AND SENTIMENT ANALYSIS IN MULTI-PARTY CONVERSATION CONTEXTS

FAST-NUCES / Dept. of AI & Data Science

Assignment 03 – Experimentation and Expansion

Group – 07

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Video – MobileNetV2+Transformer	47.22	42.94	48.33	36.14
Face – InceptionResNetV1+BiLSTM	47.22	42.94	48.33	22.61

Table 1: Assignment 2 unimodal emotion recognition results (Accuracy %)

Modality	Train Acc (%)	Dev Acc (%)	Test Acc (%)	Paper Acc (%)
Text – RoBERTa-base	81.90	65.85	70.26	69.21
Audio – Wav2Vec2-base	86.61	83.33	85.84	56.20
Video – MobileNetV2+Transformer	47.16	42.68	48.21	42.51
Face – InceptionResNetV1+BiLSTM	47.63	43.29	47.69	38.98

Table 2: Assignment 2 unimodal sentiment analysis results (Accuracy %)

3.2. Multimodal Fusion Results from Assignment 2

Configuration	Emotion Test (%)	Sentiment Test (%)	Paper Emotion (%)	Paper Sentiment (%)
Text + Audio + Video	63.50	86.76	66.29	71.76
Text + Audio + Face	63.24	86.30	66.25	71.84
Text + Audio + Face + Video (4-Modal)	64.01	87.21	66.36	72.15

Table 3: Assignment 2 multimodal fusion results (Accuracy %)

4. PROPOSED METHOD

This section presents the proposed method for conducting a gated fusion and cross domain transfer.

4.1. Research Hypothesis

Three hypotheses are tested in this assignment:

- The first hypothesis (H1) is about gated fusion. It proposes that a gated attention fusion model, which automatically learns the weight to be given to each modality for each sample, will outperform the baseline MLP with simple feature concatenation from Assignment 2. This is because the gated model can down-weight the noisy modalities, such as low-visual cues, rather than relying equally on all modalities.
- The second hypothesis (H2) relates to cross-domain transfer. It suggests that training a RoBERTa text model on the IEMOCAP dataset, and then fine-tuning it to MELD dataset, will perform better than the model that has not been trained (zero-shot). This is because the two datasets have similar emotion distributions, despite being from different domains.
- The third hypothesis (H3) relates to task complexity. It proposes that emotion recognition will be harder than sentiment analysis in both transfer learning and gated fusion model. This is due to the increased number of classes and the subtlety of emotions, and also because the dataset has an uneven distribution of the different emotions.

4.2. Method A: Gated Attention Fusion

The Assignment 2 model fuses the embeddings from the four modalities by concatenation and passing them through a fixed three-layer MLP. This assigns equal weights to all modalities regardless of their reliability on a sample-specific basis. For example, if the lighting of a video clip is dark, the face and video embeddings will be less informative, but are given the same weight as the text embedding.

Our gated attention fusion model overcomes this by first projecting the features of each modality into a shared hidden space, and then calculating a modality SoftMax gate over the modalities for each sample. In particular, for an input utterance with modalities x_{text} , x_{audio} , x_{face} , x_{video} , each is first normalized with LayerNorm and then projected into a 256-dimensional shared space, by a two-layer MLP with GELU activation. A small gate network produces a logit for each modality, and these four logits are normalized with SoftMax to form sample-specific weights that sum to one. The linear combination of the projected modalities is added to their mean, and then the result is passed through a LayerNorm and a two-layer MLP classifier. During training, a modality dropout (dropout rate 0.15) is used: for each sample, each modality is masked independently with probability 0.15 to avoid the model relying on a single modality. All samples have at least one of the modalities present. This helps the gate learn to generalize across different modality availabilities, rather than always choosing the same one.

4.3. Method B: Cross-Domain Transfer from IEMOCAP

IEMOCAP (Interactive Emotional Dyadic Motion Capture) is a popular emotion corpus collected in the studio with two actors per session. It includes text, audio and video from five sessions. We use it as an additional pretraining corpus for the text-only modality, based on the roberta-base model.

To transfer emotions, IEMOCAP emotions are mapped to MELD emotions. The emotion frustration, which is not available in MELD, is remapped to anger. We use Sessions 1-3 as training, Session 4 as development and Session 5 as test.

For sentiment transfer, IEMOCAP does not have sentiment labels. We use pseudo-sentiment labels derived from emotion labels by mapping anger, disgust, fear and sadness to negative; neutral to neutral; and joy and surprise to positive. We use the same train/dev/test split as before.

In both transfer scenarios there are three stages. Phase 1 is source pretraining: the RoBERTa model is fine-tuned on IEMOCAP for five epochs with macro-F1 calculated on the IEMOCAP development set used as the selection criterion. Phase 2 is evaluation: the IEMOCAP-pretrained model is evaluated directly on the MELD test set without any fine-tuning on MELD. Phase 3 is MELD fine-tuning: the IEMOCAP-pretrained model is then fine-tuned on the MELD training set for five epochs and evaluated on the MELD test set.

5. EXPERIMENTAL SETUP

This section discusses the ease at which the experiments can be repeated and the reason why results and model behavior however were not similar.

5.1. Hardware

All experiments in Assignment 3 were run on the same Kaggle environment as Assignment 2: two NVIDIA Tesla T4 GPUs (with 15 GB VRAM each), 4-core Intel Xeon 2.00 GHz processor, 30 GB RAM, and a 12-hour time limit per notebook. The cross-domain experiments involved downloading IEMOCAP using KaggleHub (dataset handle: sangayb/iemocap) in the same session.

5.2. Datasets

There are two datasets used in Assignment 3: MELD and IEMOCAP. For MELD, we use the 15% class-balanced subset of the dataset as used in Assignment 2, and we report the final results on the MELD test set. This includes 1,497 training utterances, 164 development utterances, and 390 test utterances for sentiment, and 1,494 training utterances, 163 development utterances and 389 test utterances for emotion after matching utterances across modalities in the gated fusion experiment. We use IEMOCAP as the pretraining dataset. We download the dataset (Sessions 1-5) via KaggleHub and create 4,341 training, 1,538

development, and 1,650 test utterances for pseudo-sentiment (and similarly for emotion after label mapping).

5.3. Hyperparameter Configuration

Hyperparameter	Gated Fusion	Cross-Domain Text (IEMOCAP)	MELD Fine-Tune
Optimizer	AdamW	AdamW	AdamW
Learning Rate	1e-3	2e-5	2e-5
Batch Size	32	16	16
Epochs	5	5	5
Hidden Dim	256	N/A (RoBERTa CLS)	N/A (RoBERTa CLS)
Dropout	0.30	0.10 (HF default)	0.10 (HF default)
Modality Dropout	0.15	N/A	N/A
Model Selection Metric	Macro-F1 (dev)	Macro-F1 (dev)	Macro-F1 (dev)
Random Seed	42	42	42

Table 4: Hyperparameter configuration for Assignment 3 experiments

5.4. Evaluation Metrics

All of the Assignment 3 experiments will report classification accuracy (to compare with the Assignment 2 and original paper), macro-F1 (to ensure balanced performance across all classes, which is important in class imbalance), and weighted-F1 (for completeness). Confusion matrices and class-specific precision/recall/F1 scores (from scikit-learn classification_report) are saved in the metrics JSON files.

6. RESULTS AND ANALYSIS

The following section presents the evaluation results obtained by reproduction of models, pretraining and fine-tuning performance, and compares with the baseline results in assignment 2.

6.1. Gated Fusion Results

Table 5 compared the gated fusion model against the flat-concatenation baseline from assignment 2 on both tasks.

Model	Emotion Acc (%)	Emotion Macro-F1	Sentiment Acc (%)	Sentiment Macro-F1
4-Modal Early Fusion	62.47	–	88.13	–
Gated Attention Fusion	60.67	0.3680	85.39	0.6285

Table 5: Gated fusion vs early fusion baseline on the MELD test set

The Gated Fusion Model didn't perform as well as the Assignment 2 baseline over both tasks. The loss of emotion accuracy was -1.8% and a decrease in sentiment accuracy of -2.74% occurred. Tables 6 present the modality weights learned and provide the most helpful diagnostic information.

Task	Text Weight	Audio Weight	Face Weight	Video Weight
Emotion (test set)	0.999794	0.000000329	0.000206	<0.0000001
Sentiment (test set)	0.999737	<0.0000001	0.000263	0.000000427

Table 6: Average modality weights from the gated attention fusion model on the MELD test set

Approximately 99.97 - 99.98% of the entire weight of the two tasks was assigned to the textual modality when the gate had fully collapsed to text. The audio, face and video modalities received weights that ranged from approximately 10^{-7} to 10^{-4} . Because of this fact, the model did not out-perform the baseline because the gated fusion acted basically as a text-only classifier with no benefits derived from the remaining three modalities.

6.2. Cross-Domain Transfer Results: Emotion

Table 7 shows the three evaluation stages for the emotion cross-domain transfer experiment.

Stage	Dataset Evaluated	Test Acc (%)	Macro-F1	Weighted-F1
IEMOCAP Pretraining	IEMOCAP Emotion	56.18	0.3761	0.5748
Zero-Shot Transfer	MELD Emotion	39.59	0.2599	0.3936
Fine-Tuned on MELD	MELD Emotion	51.93	0.4013	0.5385
MELD Text Baseline (Asgn 2)	MELD Emotion	60.93	–	–

Table 7: Cross-domain emotion transfer – IEMOCAP to MELD

When applying the pre-trained model (IEMOCAP) to the IEMOCAP dataset, the model achieves an accuracy score of 56.18%. By comparison, zero-shot transfer learning from IEMOCAP to MELD yields a much lower accuracy score of only 39.59% for MELD. This is significantly less than the MELD text baseline of 60.93%, which indicates a strong domain gap between the MELD and IEMOCAP data. After fine-tuning on MELD for five epochs, the fine-tuned model's accuracy increases to 51.93%, and its macro-F1 score increases from 0.2599 to 0.4013. Therefore, the fine-tuned transfer model is still 9 points behind the Assignment 2 MELD-only text baseline.

6.3. Cross-Domain Transfer Results: Sentiment

Stage	Dataset Evaluated	Test Acc (%)	Macro-F1	Weighted-F1
IEMOCAP Pseudo-Sentiment Pretraining	IEMOCAP Pseudo-Sent.	69.33	0.6739	0.6985
Zero-Shot Transfer	MELD Sentiment	51.03	0.5062	0.5242
Fine-Tuned on MELD	MELD Sentiment	66.67	0.6417	0.6713
MELD Text Baseline (Asgn 2)	MELD Sentiment	70.26	–	–

Table 8: Cross-domain pseudo-sentiment transfer – IEMOCAP to MELD

The pseudo-sentiment pre-training from IEMOCAP has achieved an accuracy of 69.33% in the IEMOCAP test set where we also calculated a macro-F1 score of 0.6739 means that the pseudo labels are able to be learned. The zero-shot transfer to MELD shows accuracy of 51.03% and a macro-F1 score of 0.5062, but once we fine-tune on MELD our accuracy increased from 51.03% to a total of 66.67% along with a macro-F1 score of 0.6417. Therefore, the model that we fine-tuned across domains was close to the Assignment 2 MELD-only text baseline by 3.59%.

7. DISCUSSION

This section talks about the findings, reasons why the models acted as they did and the evaluation of the hypotheses. It also describes the drawbacks of the study.

7.1. Why Gated Fusion did not improve over the baseline?

The gated fusion model converged to a trivial solution where the SoftMax gate gave the text modality a weight of ~99.97-99.98%, with close to zero weight for audio, face and video. This was because the RoBERTa text embeddings are much better and more informative, while other modalities were less informative due to being trained only on 15% of MELD. During training, the model learned that using text embeddings led to low loss, so the gate quickly biased towards text. Modality dropout (0.15) could not fix this problem because text was still present in most samples (~85%). More dropout or constraining the gate would be required. In summary, gated fusion did not help, but it does reveal that text is stronger than other modalities and SoftMax cannot balance them when one is dominant.

7.2. Cross-Domain Transfer: Outcomes and Interpretation?

Our findings on cross-domain transfer partially validate H2. IEMOCAP pretraining and subsequent fine-tuning improves transfer results over zero-shot. The emotion accuracy rises from 39.59% to 51.93%, and the sentiment accuracy from 51.03% to 66.67%, indicating that the pretraining task is a good starting point. But the models are still worse than the baseline model trained on MELD data (60.93% for emotion, 70.26% for sentiment). This is largely because of the domain difference: IEMOCAP is less natural and more controlled, while MELD is more natural, noisy and involves conversations with multiple speakers and different language styles. The difference is smaller for sentiment (3.59%) than emotion (9%). This confirms H3, showing that Sentiment can be transferred better across domains than emotion, as it involves fewer classes and boundaries.

7.3. Limitations?

There are a couple of limitations. For gated fusion, the dropout (0.15) was not large enough to prevent the model from collapsing. In addition, the gating was learned on frozen (pre-extracted) features for the different modalities, so it could not learn better representations for the modalities, and the 15% subset of MELD may not be diverse enough for multimodal learning. For cross-domain learning, only text was used since the audio/video was not aligned. The IEMOCAP sentiment labels were also approximate (not annotations), and the five epochs of fine-tuning may not have been sufficient. Lastly, both experiments used a small MELD subset (15%). The results on some emotion classes are not statistically significant due to limited samples.

8. CONCLUSION

This report described two extensions for Assignment 3 of the Assignment 2 MELD multimodal emotion and sentiment pipeline. The gated attention fusion model showed that a per-sample learned weighting of the modalities, using pre-extracted embeddings of uneven quality, will lead to a collapse of the weighting to the "best" modality. The estimated weights (text: ~99.98%, all others: <0.03%) make the failure mode clear and offer a quantitative measure to understand why the model did not perform better than simple concatenation.

The IEMOCAP cross-domain transfer experiments showed that a model pretrained on a different structure will serve as a good starting point: fine-tuning IEMOCAP-pretrained models improves over zero-shot transfer by 12-16 percentage points. But the IEMOCAP (scripted studio dyadic speech) to MELD (informal multi-party TV conversational speech) domain gap is too large to achieve better results with either fine-tuned model than with a model trained on MELD.

Overall, the experiments confirm the original paper's conclusion that text is the most informative modality for both tasks, and demonstrate that successful multimodal fusion and cross-domain transfer require either better visual and audio models, more MELD training data, or multimodal fusion strategies with explicit constraints on the sample entropy to avoid modality collapse.

9. REFERENCES

- [1] Farhadipour, A., Ranjbar, H., Chapariniya, M., Vukovic, T., Ebling, S., & Dellwo, V. (2025). Multimodal Emotion Recognition and Sentiment Analysis in Multi-Party Conversation Contexts. *arXiv:2503.06805*.
- [2] Busso, C., Bulut, M., Lee, C.-C., Kazemzadeh, A., Mower, E., Kim, S., Chang, J. N., Lee, S., & Narayanan, S. S. (2008). IEMOCAP: Interactive emotional dyadic motion capture database. *Language Resources and Evaluation*, 42(4), 335–359.
- [3] Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E., & Mihalcea, R. (2018). MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations. *arXiv:1810.02508*.

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- [4] Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv:1907.11692*.
- [5] Baevski, A., Zhou, Y., Mohamed, A., & Auli, M. (2020). wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. *Advances in Neural Information Processing Systems*, 33, 12449–12460.
- [6] Barbieri, F., Camacho-Collados, J., Neves, L., & Espinosa-Anke, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. *arXiv:2010.12421*.
- [7] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 4510–4520.